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AI in higher education: unveiling academicians' perspectives on teaching, research, and ethics in the age of ChatGPT

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ABSTRACT

This study examines the potentials and challenges of integrating conversational AI (Artificial Intelligence) tools in higher education. Specifically, the study delves into how ChatGPT is reshaping teaching, learning, research, and student evaluation while addressing ethical concerns such as academic integrity. We employ a qualitative research design based on semi-structured interviews with 12 highly accomplished academicians from North America, Asia, and Europe's higher education institutions. Our analysis reveals ChatGPT's potential to uplift educational productivity, learning outcomes, creativity, and idea generation but raises apprehensions about academic honesty, authenticity, and over-reliance. While ChatGPT can enhance educational practices such as brainstorming, analysis, and skills acquisition, its unchecked use may impede learning, critical thinking, and research quality. The study advocates for the responsible AI use in academia, underscoring educators and policymakers to establish policies guiding conversational AI ethical applications. Thus, training faculty and executing institutional policies are essential for effective and ethical integration of ChatGPT and other AI tools in higher education. Finally, this research contributes to ongoing research on AI in education by providing valuable insights that can harness potential of these transformative tools while upholding academic integrity and standards.

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ChatGPT; conversational AI; generative AI; artificial intelligence; education; learning; teaching; research

Introduction

Stephen Hawking, a prominent scientist, expressed his belief that AI (artificial intelligence) “*will either be the best thing that’s ever happened to us, or it will be the worst thing*” (Higgins, 2018). This statement highlights the polarizing impact of AI on society and underscores the crucial need for responsible AI design and implementation. Recent technological advancements have significantly boosted enthusiasm for conversational AI. A notable example is ChatGPT, a versatile chatbot developed by OpenAI that utilizes the powerful GPT-3 LLM (large language model). ChatGPT’s success is marked by its quick adoption rate, amassing over 1 million users within five days and 100 million users within three months. Its growth significantly outpaces previous digital and internet technologies such as mobile phones, Facebook, TikTok, and Instagram, resulting in a valuation exceeding \$20 billion (Garfinkle, 2023).

Modern advancements like the LLM GPT-3 differ from earlier methods because they use foundational AI models trained on large datasets through self-supervision, rather than relying on supervised learning with explicit task labels (Bommasani et al., 2022). This unique characteristic empowers

ChatGPT to be versatile and adaptable, enabling it to handle various downstream tasks based on user prompts and deliver authoritative human-like responses across diverse topics and tasks (Rudolph et al., 2023).

With the fast-paced progress in modern AI, especially in generative AI, there's a rising interest in applying AI systems to education, with the goal of transforming classroom teaching and scientific research. The growing enthusiasm for integrating ChatGPT into educational contexts is apparent in discussions on social media platforms and the increasing number of research dedicated to this topic (Choi et al., 2023; Ibrahim et al., 2023; Qadir, 2023).

This enthusiasm stems from ChatGPT's diverse range of applications, which include content generation, personalized responses, and multilingual translations.

Integrating AI into education has the potential to significantly improve educational outcomes, student engagement, and motivation (Wang et al., 2023). Aydin and Karaarslan (2023) emphasize that generative AI tools like ChatGPT can streamline academic workflows, enhancing efficiency. Through immediate feedback and assistance, ChatGPT facilitates the learning process, leading to better knowledge acquisition and retention (Rudolph et al., 2023). It also simplifies evaluation by providing accurate and timely feedback to teachers on student performance. Additionally, ChatGPT's real-time coaching and support benefit students with special learning needs, such as non-native language speakers, helping them overcome language barriers. These capabilities demonstrate ChatGPT's potential to support diverse learners and foster inclusive educational environments.

Beyond teaching and learning, generative AI is poised to disrupt academic research practices. While analytical and creative thinking remain vital, ChatGPT offers researchers extensive access to textual resources and information, revolutionizing the academic landscape by streamlining the writing process. This accessibility enhances the efficiency of academic writing, information dissemination in research, and the creation of course materials and assessments. Conversely, generative AI models are known to present issues such as "hallucination," whereby inaccurate information is produced and presented as true and valid. Therefore, this gives rise to problems related to originality and ethics in research, and educators worry about the accuracy of such AI tools (Rudolph et al., 2023). Concerns have also been raised about how AI will affect job opportunities in various fields, in particular education (Ore & Sposato, 2022).

While some argue for a complete ban on ChatGPT due to concerns surrounding its adoption (Baron, 2023; A. Williams, 2023), prohibiting its use is not a realistic response for various reasons. *Firstly*, banning ChatGPT would hinder genuine educational advancement by impeding its potential to enhance traditional teaching methods, and provide personalized assistance and real-time information to improve the learning experience. *Secondly*, a ban would stifle progress in the field of AI and limit opportunities for innovation, as continued research and development are crucial for refining AI tools like ChatGPT. *Additionally*, the lack of foolproof methods to detect its use, despite the emergence of AI-writing detectors, undermines the feasibility of a ban. Instead, the emerging community practice is leaning towards acceptance and advocating for ethical and value-adding use of such tools (Qadir, 2023).

Proponents of the use of such tools in education (Hargreaves, 2023; Tlili et al., 2023) argue that it is important to educate students on how to make efficient use of ChatGPT. Introducing these tools to students with caution is considered the best approach to help them enhance their skills, knowledge, and overall productivity. They advocate for teaching students how to access, browse, and effectively utilize such tools, as well as how to formulate questions that yield accurate responses. Instructors can play a crucial role in ensuring that ChatGPT is used appropriately by actively exploring its features and limitations to gain a comprehensive understanding of its functionality. Promoting responsible guidelines and usage can address concerns while leveraging the benefits of ChatGPT in educational settings, fostering innovation, and driving improved learning outcomes.

In this study, we explore how the use of ChatGPT in educational institutions has the potential to drastically affect research methods, student assessment, and classroom instruction. We delve into interview data based on first-hand accounts of how academicians view ChatGPT. The findings of

this research study can lay the groundwork for enhancing learning outcomes, facilitating the evaluation of instructional content, and streamlining research processes. We also explore the challenges related to the ethical usage of ChatGPT, problems regarding users' privacy, and ethics that need to be addressed.

The main goal of this study is to explore instructors' views on the following questions:

- RQ1. (ChatGPT and traditional pedagogy):** How does the adoption of ChatGPT affect traditional teaching and pedagogy?
- RQ2. (ChatGPT and student outcomes):** How does the use of ChatGPT impact the learning outcomes of students?
- RQ3. (ChatGPT and research):** What are the implications of integrating ChatGPT in scholarly research?
- RQ4. (Ethics of ChatGPT's use in academia):** What are the ethical issues pertaining to ChatGPT's use in academia?

Background and related work

ChatGPT functionalities

Through the platform, ChatGPT offers a wide range of applications related to natural language processing and generation (Lund et al., 2023). It can effectively answer questions, from basic to complex, on virtually any subject. Additionally, it excels at text generation tasks such as content creation, creative writing, summarization, and multilingual translation including languages like Finnish, Urdu, Spanish, and Arabic. Users can leverage ChatGPT to generate resumes, write essays, and solve assignments. The tool accurately performs rewriting or paraphrasing in specific voices or genres. Moreover, it facilitates conceptualizing and developing true or false questions, multiple-choice questions, case studies, and quizzes (Gonsalves, 2023; Kiyak et al., 2024; Tlili et al., 2023). ChatGPT is also capable of generating programming codes for various functions in almost any programming language. In educational settings, it assists instructors in developing lesson plans, course outlines, unit plans, exam questions, reading lists, and providing responses to student assessments (Tlili et al., 2023). Moreover, in the realm of scholarly research, ChatGPT proves highly valuable for study design, literature reviews, and composing structured abstracts, introductions, results, and discussions on diverse topics (Dowling & Lucey, 2023; Haman & Školník, 2023; Macdonald et al., 2023).

ChatGPT and traditional pedagogy

Whenever a disruptive technology is proposed, it takes some time for the community and the field to adjust. Numerous reservations were raised within the education sector when novel inventions such as calculators, televisions, personal computers, video games, and the internet became popular and accessible to students (Chou et al., 2005; Gentile et al., 2004; Suydam, 1979; P. A. Williams et al., 1982). For instance, the widespread adoption of calculators in the 1970s led to debates about the strengths and weaknesses of using calculators in the classroom and in general learning mathematics. Some researchers stressed the significance of calculators as they can make tedious calculations quickly without making mistakes and allow the learner to spend more time refining mathematical concepts. On the other hand, there were worries related to over-reliance on calculators that the students would become overly reliant on calculators, and it might deteriorate their mathematical concepts and skills, ultimately impacting their learning and performance (Suydam, 1979). In response, many higher education institutes opted for policies and guidelines to limit or altogether ban calculators' usage in class or during an examination, while many others allowed and promoted their use as a staple for learning mental mathematics skills (Bukowski, 1975; Kissane, 2000).

In exploring instructors' perceptions of ChatGPT within higher education across three continents, it's crucial to contextualize these views within the broader history of educational technology. It's worth noting that in the past, several technologies were hailed as revolutionary but ultimately fell short of the anticipated impact (Winner, 2009). For example, distance learning courses broadcasted on television and massive open online courses (MOOCs) were expected to revolutionize education, potentially rendering traditional universities and teachers obsolete. However, as Toyama argued in his book "Geek Heresy" (Toyama, 2015), history has shown that technology alone cannot bring about significant social change in education. It necessitates the presence of appropriate analog complements, such as well-trained teachers, sound policies, and guidelines, for technology to truly be effective. This historical perspective underscores the importance of examining the nuanced reactions to ChatGPT among educators, which may reflect similar patterns of technological adoption and adaptation.

ChatGPT and student outcomes

The majority of recent research on ChatGPT use in education has been published as editorial pieces or has concentrated on experimental design from a methodological standpoint. According to the majority of studies ChatGPT is a unique, practical, and beneficial tool for students (Gilson et al., 2023; Halaweh, 2023; Kasneci et al., 2023; Rudolph et al., 2023; Zhai, 2022). From the standpoint of the learners, difficult and complex subjects in a variety of fields (such as finance, medicine, software engineering, chemistry, and law) can be simply elaborated for group projects, independent study, and test preparation (Choi et al., 2023; Gilson et al., 2023; Tlili et al., 2023).

ChatGPT is also regarded as a highly useful tool in clinical decision-making (Gilson et al., 2023; Kung et al., 2023). For medical students, ChatGPT is considered an instructional tool for selfstudy and a quick reference tool bearing a high degree of accuracy when answering questions (Nisar & Aslam, 2023). A number of experimental studies attempted professional examinations, like the University of Minnesota Law School examination, the United States Bar examination, and the United States Medical Licensing Examination (USMLE). According to these tests, the responses produced average passing scores and demonstrated ChatGPT's ability to contextualize and provide pertinent responses (Choi et al., 2023; Kung et al., 2023). However, other research indicates that ChatGPT may have a detrimental effect on learners' motivation and the skills that are needed as a result of learning such as innovation capabilities and critical thinking (Kasneci et al., 2023; Tlili et al., 2023).

Research & ethics in the context of ChatGPT

From the standpoint of academic research, ChatGPT exudes exceptional potential. Prior to the release of GPT-3, Compared to any other LLM that existed prior to the launch of GPT-3 (e.g. BERT, RoBERTa, T5, and XLNet etc.), ChatGPT outperformed others in terms of answer accuracy, topic variety, and, most importantly, the conversational interface, which allows anyone to use the platform and interpret the results with ease. The tool has several advantages in various research domains, including but not limited to medicine, law, finance, information technology, and business, that could potentially accelerate the scientific process. Several studies have experimented with writing sections or entire research papers on the platforms (Aydin & Karaarslan, 2023; Dowling & Lucey, 2023; Tlili et al., 2023). Aydin and Karaarslan (2023) used the platform to write a review on "Digital Twins in Healthcare." Their investigation revealed that using tools like ChatGPT for scholarly research will most likely encourage knowledge compilation and effective presentation of research findings. Furthermore, the tool is useful for generating new ideas, designing studies, identifying and analyzing data, and drafting manuscript content (Macdonald et al., 2023).

Several studies found that LLM-generated content can be factually inaccurate, incorrect, biased, or otherwise inappropriate, posing ethical and social risks (Kitamura, 2023; Mhlanga, 2023; Tokayev, 2023; van Dis et al., 2023). There have been several concerns raised about cheating, plagiarism,

ethics, and copyrights in relation to ChatGPT-generated content. The study by Yeadon et al.'s (2022) study viewed ChatGPT as a risk to essay writing, which is a crucial type of evaluation that is frequently employed in a variety of domains (Yeadon et al., 2022). The uniqueness and referencing of the material produced by ChatGPT has also drawn criticism. Numerous studies show that ChatGPT is unable to provide references or sources to support the veracity of the text it produces (Haluza & Jungwirth, 2023; Nisar & Aslam, 2023). This restriction makes it more difficult for the user to find in-depth explanations of the generated content. Because of the frequently low similarity rates, typical plagiarism detection technologies have difficulty detecting instances of plagiarism in text generated by ChatGPT, despite the tool's ability to produce clear and succinct writing.(Aydin & Kararslan, 2023; Gao et al., 2022). To combat AI-generated content, detection engines have been created. A few studies concluded that the platform's content was fake, inaccurate, deceptive, or prejudiced (Tlili et al., 2023). Furthermore, in many cases, ChatGPT fails to account user emotions and contextualize information. These constraints suggest that human understanding is required at all stages of research to ensure precision if this platform is implemented (Dowling & Lucey, 2023; Haman & Školník, 2023; Macdonald et al., 2023).

Methods

The present study adopts a qualitative research design, employing semi-structured interviews as the primary data collection method. Semi-structured interviews are deemed highly suitable for exploratory investigations aimed at gaining initial insights into a phenomenon (Galletta, 2013; Mayring, 2014). Widely utilized in research, these interviews help in comprehending and establishing a theoretical groundwork for emerging topics (Kallio et al., 2016).

In the present study, we interviewed academics who are currently working in three different continents, i.e. North America, Europe, and Asia. The study participants were faculty members from the United States ($n = 4$), Oman ($n = 4$), Finland ($n = 2$), and 1 participant each from Sweden and the United Kingdom. Three independent researchers sent emails to several potential participants and inquired about their current understanding of using ChatGPT, as well as solicited their consent to participate in the interview. Only those persons were included who had experience using ChatGPT and showed their consent to participate in the interview voluntarily. All data collected were de-identified to protect the privacy of the participants. Each participant was interviewed individually in person or through an online meeting. Before starting the interview, we explained to them the purpose of the study, the confidentiality of their responses, and the time required for the interview, reiterated their right to voluntary participation, and told them that the interview would be recorded for transcription purposes. We also asked the respondents if they had any questions or concerns regarding the discussion or the topic of the study. After completing all the aforementioned

Table 1. Demographic profiles of respondents.

Factor		Frequency	Percentage
Gender	Male	09	75.0
	Female	03	25.0
Age	26–35 years	01	8.3
	36–45 years	07	58.3
	46 years & above	04	33.3
Education	Masters	01	08.3
	PhD	11	91.7
Academia	3 years or less	02	16.6
Experience	3–6 years	01	08.3
	6–9 years	09	75.0
Continent	North America	04	33.3
	Europe	04	33.3
	Asia	04	33.3
		N = 12	100

requirements, we conducted and recorded all the interviews. [Table 1](#) shows the demographic profile of the respondents. The table indicates that there were 09 male participants and 03 female participants. All the participants hold a doctorate degree and have been working in academia for more than three years. The participation from the three continents remains equal.

The interview questions consisted of a mix of closed and open-ended questions about their experience using ChatGPT, including any challenges they have encountered, how they have incorporated the model into their work, and any suggestions for improvements. After completing and recording the interviews, we transcribed the interview recording using two transcription tools, Otter and Temi. Three independent researchers then manually verified the transcription to ensure error-free data. During the verification process, the researchers corrected the errors and omissions made by the transcription software. Interview transcripts were examined using thematic analysis to identify patterns and themes in the data (Maguire & Delahunt, 2017; Mayring, 2014).

Results & findings

The interview data was analyzed using a thematic analysis. In the thematic analysis we have used coding and grouped the words and patterns in five distinct categories and labeled each of them. Through the thematic analysis of the interviews, we discovered that the concerns expressed by the participants related to five main themes: teaching, learning, assessment, research, and ethics (see [Table 2](#)). Furthermore, the coding approach supported in extracting sub-themes that were organized based on common words and patterns. We extracted 23 sub-themes that include 6 subthemes for teaching, 5 sub-themes for learning, 2 sub-themes for assessment, 4 sub-themes for research and 6 sub-themes for ethics.

The following paragraphs discuss each theme and sub-themes in detail.

Teaching

Overall, participants had a favorable impression of utilizing ChatGPT in designing and arranging lectures and tutorials. This was the case regardless of whether the process was conducted face-toface or online. They emphasized that the tool's benefits extend beyond saving time to enabling educators to concentrate on other essential aspects of teaching. For instance, one of the participants echoed that "it saves time and enables them to concentrate on more important areas of the educational experience."

The study's results show that individualized instruction is another possible benefit of ChatGPT.

This is especially helpful for students who have trouble learning. Incorporating AI tools such as ChatGPT can positively impact the overall quality of the educational experience; however, teachers must be trained to use them effectively. Participants pointed out that to successfully integrate AI tools such as ChatGPT into the overall learning experience, a number of factors need to be considered. For example, a participant stated that:

"It is necessary to provide teachers with training on how to use them. This necessitates not only having access to the appropriate tools but also having the knowledge and expertise necessary to make successful use of those tools."

The study results further show that when AI tools are used in educational settings, the traditional way of teaching needs to be changed. A participant noted that:

"This requires the implementation of active learning practices that create new information for the future, requiring a complete overhaul of the educational system, and an emphasis on educating teachers to use AI tools effectively."

Furthermore, ChatGPT could be a helpful tool in redesigning the curriculum, even though its use in coming up with new ideas was seen as both positive and negative. For example, a participant stated the following:

Table 2. Major themes and sub-themes identified.

Theme	Sub-themes
Teaching	1. Lecture design & planning (+) 2. Tutorial design (+) 3. Personalized teaching, especially with those with learning disabilities (+) 4. Teachers to be trained to use AI tools 5. Curriculum redesign 6. Idea generation (+–)
Learning	1. Productivity enhancement, time saving / effort reduction (+) 2. Creativity (+–) 3. Skill acquisition (+–) 4. Superficial understanding (–) 5. Customized learning
Assessment	1. Restructuring Assessments • Increased in-class assessments • Oral assessments • Applied assessments (Problem-based, case studies) • Integration of AI in assessments (e.g. context-based) 2. Teacher training • Restructuring assessments • Utilization of AI for effective/relevant assignments • Cognizance of AI versus human-generated work
Research	1. Writing • Improve content writing (+) • Negatively impact writing skills, lack of writing practice (–) • Text dumping (–) • Accelerate the writing process 2. Idea generation • Provides alternative research ideas (e.g. seeding ideas) (+) • Formulating research questions and abstracts (+) • New variables and constructs (+) • Limited scope of ideas & thinking ability (–) 3. Analysis, reflection & creativity • Critical thinking (–) • Lack of analysis (–) • Inhibit human reflection (–) • Increase or decrease creativity (+–) 4. Research process • Undermine meaningful research process (–) • A shallow understanding of research problems & methods (–) • Lack of rigor (unable to validate and verify) (–) • Need to draft clear policies regarding ChatGPT / AI tools • Negatively impact quality/originality of research • Attributing/acknowledging its usage
Ethics	1. Copyright issues • Intellectual Property Rights Issues 2. Authenticity and reliability (–) • Source of information is unknown • Lack of any citation • Not everything produced by AI is accurate & empirically sound 3. Limited scope of AI information retrieval and algorithm bias • Produces what the algorithm is trained for or limited sources of data • Inaccurate information may keep on rolling and magnified 4. AI safety, AI regulation, AI watermarking 5. Academic integrity (–)

(Continued)

Table 2. Continued.

Theme	Sub-themes
	• Dishonesty • Plagiarism
	6. Information reproduction & ethics (+–)
	• Can help support basic information collection to speed up work (+) • Ready-made boilerplate solutions/ethics (–) • Bias (e.g. anti-Islamic or Islamophobic content) (–)

“Assignments need to be restructured to be more immersive and hands-on in order to get the most out of the benefits that AI technologies may bring to the learning process”.

AI tools such as ChatGPT can also contribute to a more inclusive learning environment, which assists all students in performing at the same level. This is especially helpful for teachers and students who have disabilities such as blindness or other conditions that make it difficult for them to learn new things. An interviewee stated the following:

“I would see ChatGPT as being a win-win for a lot of students who may struggle due to language impairments. Having something that could help them, would be very beneficial.”

Even though AI tools could be helpful, some participants said they were afraid that using them would lead to a shallow understanding of the topic and stop people from coming up with new ideas.

Learning

Learning is at a crossroads. This research aimed to investigate how using ChatGPT influences the learning process. The individuals who participated in the study presented various points of view regarding the impact that ChatGPT has had on the educational system, identifying both positive and negative aspects. Generally, participants held a positive view of ChatGPT in facilitating learning. For example, a participant stated that:

“ChatGPT could be used as a thought starter or a springboard to begin their task, which would improve the overall learning experience.”

It was also mentioned that using ChatGPT could help students be more productive in their assignments by making it take less time and effort to complete them. A participant mentioned that ChatGPT “can be used for sentence construction far better than Grammarly.” The participants believe that the tool may be helpful for both students and teachers since it may make it easier for either group to concentrate on other critically important aspects of the educational experience. For example, ChatGPT might be used to automate specific tasks like grading, giving up more time for instructors to engage in more personalized conversations with students. Students could also use ChatGPT to check their work for mistakes, saving them time for proofreading and editing. A participant stated the following:

“using these AI tools will amplify the skills that we do have. You know, I mean, that’s the whole point of, you know, a word processor or, you know, writing code, you know, it’s like you’re extending your abilities. So this will extend people’s abilities.”

Results from the interviews also reveal that using ChatGPT could be a beneficial tool for facilitating individualized education. Participants observed that the tool might be utilized to provide personalized feedback and ideas for development, enabling students to learn at their own speed and difficulty level. For example, ChatGPT could be used to determine the areas in which students require additional support and then deliver specifically tailored feedback to address these areas. However, the implementation of ChatGPT needs to be meticulously planned out and reviewed before it can be considered to be in accordance with the learning objectives and core values of

the educational institution. Despite the benefits, participants raised their concerns about ChatGPT's role in learning. An interview participant expressed a widespread problem where many students seldom engage in deeper thinking and want quick and easy solutions to learning, and that AI tools are set to amplify this problem. Several participants were concerned that the tool could hinder the student's ability to enhance their writing and data analysis skills. They stated that relying too much on ChatGPT could reduce the requirement for critical thinking and analysis, both of which are necessary abilities for ongoing education. This is due to the fact that ChatGPT may offer simple solutions to difficult situations, hence reducing the need for creative thinking. It is crucial to remember that ChatGPT's effect on creative thinking may vary depending on how the tool is employed and the educational objectives that are aimed to be accomplished. A participant stated the following:

"If a student, for example, submits a written essay to you, and it's a very good essay, then you interrogate them on it, and you drill down into certain aspects of it. And you say, Well, can you explain this in more depth? And why and why and why and why I think students will struggle, because they'll only have that superficial surface learning. I think that's, that's a problem. That's a potential problem. But then again, hey, that's what students want these days. They don't want to learn anything with any depth of knowledge. They want the quick and easy way and they look for the quick and easy way. And now this artificial intelligence software is just providing them with that."

It was also revealed that using ChatGPT could result in students having only a surface-level comprehension of the material because it gives students information but does not instruct them on assessing what they read critically. A participant stated that:

"The use of ChatGPT would reduce the requirement for critical thinking and analysis, which would in turn make it easier for students to write and evaluate data. Also, ChatGPT should be used as a tool for augmenting the learning process rather than replacing it entirely as a means of education."

Assessment

The introduction of ChatGPT into the learning process necessitates reorganizing the evaluation process to align with the changing educational landscape. The participants came up with several different suggestions for how the evaluation procedure may be reorganized in order to incorporate the use of ChatGPT. This includes more testing done in the classroom, orally, based on problems or case studies, and testing done in the subject matter context. These strategies may aim to improve alignment between the learning process and the evaluation process, as well as to encourage authentic learning as a means of education. A participant stated the following:

"I think that we're gonna need to revamp the curriculum and we're gonna have to stop and do a lot of educating with teachers."

Another participant noted that:

"We'll have to be less lazy in our assessment procedures. We would likely push assignments and tasks to become more concrete and ongoing in terms of how they're assessed."

Results also point to the need for educators to receive training on integrating ChatGPT into their instruction and evaluation methods before they can make good use of the platform in the classroom. Participants emphasized to educate academics on reorganizing grading systems, using AI to create more efficient and pertinent assignments and understanding the distinctions between work generated by humans and work generated by AI.

Research

Participants discussed various elements of using ChatGPT for academic purposes during the interviews. In terms of one's ability to write, the use of ChatGPT can have both beneficial and detrimental

effects. A potential advantage of using ChatGPT is that it can speed up the writing process and save students' and teachers' time.

The individuals who took part in the research agreed that ChatGPT has the potential to enhance content production by providing comments and suggestions for further development.

Furthermore, the employment of ChatGPT has the potential to facilitate the production of new ideas. Participants mentioned that ChatGPT can offer alternate research ideas, sometimes known as "seeding ideas," and help in the process of creating research questions and abstracts. Moreover, additional variables and constructs that need to be considered can be generated using ChatGPT.

Several participants voiced concerns about the current tool's limited idea pool and the possibility that it will inhibit their capacity to think creatively. For example, a concerned participant stated that ChatGPT could have:

"A detrimental impact on writing abilities since it would reduce the need for practice and lead to text dumping, both of which could potentially lower the quality of the writing."

Participants also voiced concerns about the effect that ChatGPT would have on research-related analytical and critical thinking, given that the technology has the potential to stifle human thought and restrict analysis. There is a risk of undermining originality by simplifying previously tricky challenges. Moreover, the participants stated that ChatGPT could hinder the meaningful research process since it encourages a superficial awareness of research topics and methodologies.

Ethics

The introduction of ChatGPT in educational settings has positive and negative repercussions for the reproduction of information and ethical behavior. One of the benefits of utilizing this tool is that it can make the process of accumulating fundamental pieces of information easier and speed up work. Using pre-made templates, however, raises the possibility of unethical problems such as academic dishonesty and plagiarism. When it comes to deciding whether or not to deploy ChatGPT in educational settings, several important ethical questions need to be answered.

Incorporating ChatGPT into the education system necessitates reforming the testing procedure to adapt to the shifting conditions of the educational environment. In-class assessments, oral assessments, problem-based or case study assessments, artificial intelligence-based assessments, and context-based assessments are some ideas participants have proposed for reorganizing the evaluation process to incorporate ChatGPT. Other ideas include context-based assessments and oral assessments. The purpose of these tactics is to encourage learning that is more real and to increase alignment between the various learning practices.

By using ChatGPT, participants expressed their uneasiness with the likelihood of dishonesty and plagiarism occurring. This raises questions regarding the originality of the submitted work and ownership of the work, which may have legal repercussions. Participants also drew attention to the threat posed by students who outsource their work to third parties via the internet, which lowers the perceived worth of degrees and reduces the efficiency of the educational system. Because of this, students need to be made aware of the ethical concerns of utilizing ChatGPT, and educational institutions need to hold students accountable for the work produced using the technology.

In several educational contexts, there is concern over the legitimacy of the information that ChatGPT creates. It was brought to the participants' attention that while using ChatGPT, the origin of the information is not always transparent, and there is an absence of citations for the work that is produced. Hence, not everything AI generates is empirically sound or correct, which might lead to problems regarding the authenticity and dependability of the provided information (Figure 1).

Other smaller themes that emerged from the interview data, included regulation of AI and watermarking of AI-generated content to safeguard the ownership and its validity. Concerns have been expressed about the possible misuse of ChatGPT, which could put academic integrity and research at risk.

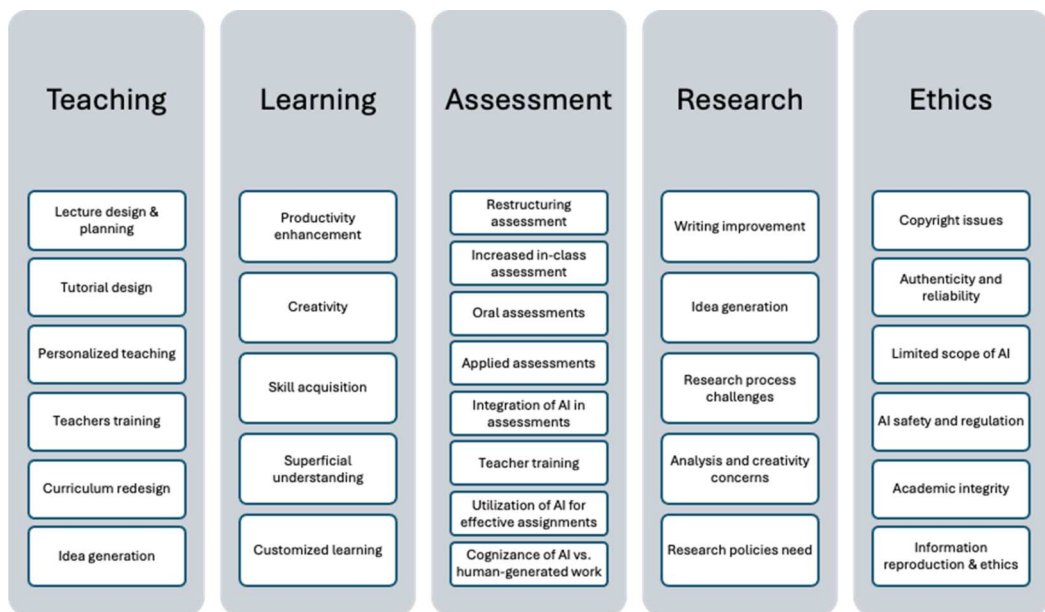


Figure 1. Overview of major themes.

Discussion

The present study sets to explore the benefits and challenges of integrating ChatGPT in academia by conducting in-depth interviews with highly accomplished academicians across three continents. This study contributes to the growing body of research on the effects of ChatGPT in academia, what academicians think about its benefits and repercussions, and how it can be channelized to achieve the greater good for the academic community. The present research enhances academicians' understanding of the use of ChatGPT and contributes to the body of knowledge in several ways.

First, this research particularly delineates the utilization of ChatGPT in teaching, research, and student assessment and finds that the use of ChatGPT is mainly subjective, which may go either way. It may enhance the efficiency and effectiveness of academic tasks or hinder the users' capabilities if used without contemplation. The study highlights that ChatGPT can be an excellent helping hand when brainstorming ideas for teaching and research, further extending the views of Macdonald et al. (2023). However, our study digs deep into the potential avenues where ChatGPT can be used for idea generation. For instance, our study suggests that teachers may use ChatGPT to generate topics for in-class discussion and student assessment, i.e. assignments and projects, articulating state-of-the-art topics for assigned courses, or developing new courses. Likewise, educators may utilize ChatGPT to develop tailored lesson plans for students with special needs (e.g. dyslexia) to improve their comprehension and engagement.

While using ChatGPT for teaching and evaluation, our study emphasizes the need to coach students on using it positively and productively rather than asking them to complete the assigned tasks. There is a high chance that students may use it to complete the assigned tasks. In this regard, Rudolph et al. (2023) assert that the advent of ChatGPT may end the traditional modes of assessments. Similarly, Zhai (2022) experimented by writing an article with ChatGPT and argued that students may do the same to complete their assignments.

Second, ChatGPT may enhance the efficiency and productivity of educators by assisting them in completing mundane tasks such as automated administrative work, streamlining documentation, and drafting emails. These findings support what Aydin and Karaarslan (2023) suggest for using assistive tools to enhance productivity. However, relying too much on ChatGPT to complete academic

and research activities might limit users' critical thinking, reflection, and creativity skills. These findings are consistent with recent literature, which recommends against utilizing ChatGPT since the generated information may be inaccurate, negatively influencing cognitive ability (Halaweh, 2023; Tlili et al., 2023).

Third, the present study's findings underscore that the AI tools support academicians to regularly enhance their skills and capabilities to meet the evolving needs of the education industry. For instance, the diligent use of assistive AI tools by educators can foster continuous professional development, collaborative learning, and virtual mentorship, ultimately enhancing their pedagogical expertise. However, using ChatGPT to complete tasks without due diligence may hinder users' skill acquisition as the users will get the outcome without adequately understanding the process. These findings complement the recent arguments of Tlili et al. (2023), who asserted that ChatGPT might negatively impact the skills and capabilities of the users. In a similar vein, the present study highlights concern regarding the credibility, authenticity, and accuracy of the information produced by ChatGPT, as there is no mechanism to validate if the information generated by ChatGPT comes from valid empirical findings. These findings support Nisar and Aslam's (2023) assertions regarding the trustworthiness of information retrieved through ChatGPT.

Finally, the present study also finds potential issues pertaining to ethics and integrity that may emerge due to the use of ChatGPT in academic settings. For instance, the lack of rigor in evaluating and verifying the information that ChatGPT generates can undermine the research quality and originality. Therefore, it is necessary to develop explicit policies regulating the usage of ChatGPT and other AI tools in educational environments. This includes recognizing its application and assigning credit for it to the results of research to foster transparency and academic integrity. The usage of ChatGPT also raises questions regarding the academic integrity of the institution. The potential to misuse ChatGPT in educational settings and produce material that violates copyright and intellectual property rights is at the heart of the ethical concerns raised by its use in these settings. Educational entities should set rules and processes to guarantee that ChatGPT is utilized in a manner that is both ethical and compliant with the laws governing intellectual property and copyright.

While employing ChatGPT, additional problems include the possibility of algorithmic bias and a restricted information retrieval scope. Because the information that is created is based on the trained algorithm and the sources of data that are available, it is possible for there to be mistakes and biases in the results. AI based algorithms can also produce biased information, such as antislamic or Islamophobic content. Educational institutions must develop policies that encourage diversity and inclusivity in applying artificial intelligence tools. It is essential for the development of policies and for enhancing transparency to have a solid understanding of how AI tools function. These findings are consistent with what Yeadon et al. (2022) highlighted earlier about the unethical use of such assistive AI tools in research and teaching. Similarly, Mhlanga (2023) suggested that the ethical use of ChatGPT in academia is very complex and needs more research attention. The current study fulfills this recommended research gap. By addressing several research gaps, the present study broadens the body of knowledge as well as it helps in setting future research direction to further the literature on the uses of assistive AI tools in academia and outlines five different avenues for future research i.e. teaching, research, student assessment, learning, and ethics by documenting several subthemes.

Practical implications

The current study offers several implications for teachers, researchers, students, academic institutions, regulatory bodies, and publishers. The interview data has shown that there are several complex and contrasting issues associated with the usage of ChatGPT in academia. The utilization of ChatGPT in academic settings comes with various positive outcomes as well as obstacles. As a result, several recommendations and policy implications can be made about how the platform may be used effectively in academia. We offer the following key recommendations:

- I *Teachers integrate ChatGPT or such assistive AI tools in effectively delivering the content.* Teachers may take help from AI tools to design and plan the content and use ChatGPT during the lecture to generate discussion ideas. Enabling students to use such tools during the class may enhance their ability to comprehend complex phenomena interactively.
- II *Assistive AI tools be used to design class assessments, including assignments, projects, and exam papers.* However, the use of ChatGPT in evaluations must include the human teacher in the loop and be restricted to circumstances in which it fosters authentic learning and is congruent with the learning goals.
- III *Rules and Guidelines for acceptable use.* Educational institutions should adopt rules and guidelines for the ethical use of ChatGPT to guarantee that it is used in an acceptable and ethical manner. The use of ChatGPT in evaluations needs to be governed by explicit policies and procedures. The parameters of use, the penalties for misuse, and the significance of academic integrity should all be outlined in policy documents.
- IV *Responsible integration of AI in Education.* Academic institutions should strive to embrace AI technologies by cautiously monitoring the novel developments of AI tools in educational settings, such as ChatGPT, to ensure that best practices are being followed. This will assist institutions in keeping abreast of the most recent research and ensuring that their rules and guidelines remain applicable and valuable in the future. In addition, academic institutions should work together with AI researchers and developers to promote the responsible and productive application of AI tools in educational settings.
- V *Importance of Training and Capacity Building.* Educational institutions should also offer instructors the training and assistance necessary to use ChatGPT in their classrooms effectively. This would help teachers to serve their students better. Studies have shown that optimal utilization of internet resources requires both internet access and motivation and vital skills to access those resources (Khan et al., 2020). AI training should also be provided to students to utilize the technology optimally and ethically effectively.

In summary, ChatGPT has the potential to completely transform how academic settings approach learning, research, and evaluation of student performance. Nonetheless, its use must be governed by clear regulations and guidelines that encourage ethical behavior, authentic learning, and academic integrity. Educational institutions can ensure the responsible and productive use of ChatGPT in academic environments by putting into action the recommendations and policy implications presented above.

Limitations of the study and future research direction

The current study is exploratory as it tries to understand an emerging phenomenon. However, it provides no quantitative evidence on using ChatGPT in academia. Future research may follow a survey approach to describe the phenomena in further detail. Our study may work as a stepping stone to building further research on this emerging topic. Similarly, our study only managed to conduct interviews of the faculty members, whereas students, researchers, and institutional decision-makers may also be interviewed for further investigation. ChatGPT is still in its early phases of development and may undergo significant changes in the coming months. Future research may take it as an opportunity to suggest what changes can be made to make it more valuable for academia and research. Though our study offers considerable implications for the relevant stakeholders, a single study can never encapsulate all the aspects of such a massive and transformative technology.

Future research should explore the expanding influence of social media as a platform for education, where independent content creators disseminate knowledge and foster skills development. Given the growing importance of generative AI technologies in social media, it is important for future research to explore how these platforms promote formal and informal learning. Platforms such as YouTube provide a powerful means for academics and professionals to share knowledge

to a broad global audience, as they function without the limitations imposed by institutions. Nevertheless, this autonomy also gives rise to apprehensions surrounding the ethical utilization of generative AI technologies. It is imperative for future research to provide guidance to regulatory agencies on upholding ethical standards in these contexts without impeding innovation.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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Appendix A. ChatGPT interview questions

- a. How familiar are you with ChatGPT and its features and capabilities? (*screening question*)
- b. Do you think ChatGPT is a useful tool for teaching and learning in education?
- c. In coming years, how will ChatGPT impact teaching?
- d. How do you think ChatGPT could be used to support students with different learning needs or abilities? (*e.g. learners with special needs and non-native speakers*)
- e. How will ChatGPT impact student assessment?
- f. Do you think ChatGPT can be a useful tool for scholarly research? If so, how would you incorporate it for your research purposes?
- g. In your opinion, how will ChatGPT impact academic research in coming years?
- h. In your opinion, what are the ethical and privacy considerations of using ChatGPT in education?
 - i. Do you think ChatGPT could potentially be misused or have negative consequences in education?
 - j. Do you have any other concerns or reservations about using ChatGPT in education?
- k. In what ways do you think ChatGPT could be improved or enhanced for use in educational settings?
- l. How would ChatGPT impact skill acquisition and creative thinking?
- m. Do you think ChatGPT might impact the role of teachers in the future?
- n. How do you think ChatGPT could potentially impact the job market or employment opportunities for teachers in the future?
- o. Do you have any final thoughts or opinions on ChatGPT?